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LITERATURE SURVEY

ON

CROSS-SPECTRAL DSMAC

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INTRODUCTION

What is DSMAC?

- DSMAC (Digital Scene Matching Area Correlation) is a vision-based navigation technique that estimates a vehicle's position by matching onboard sensor imagery with reference maps.
- It is widely used in GPS-denied environments where satellite navigation is unavailable or unreliable.
- DSMAC compares observed scene features with stored terrain or scene information to determine accurate localization.

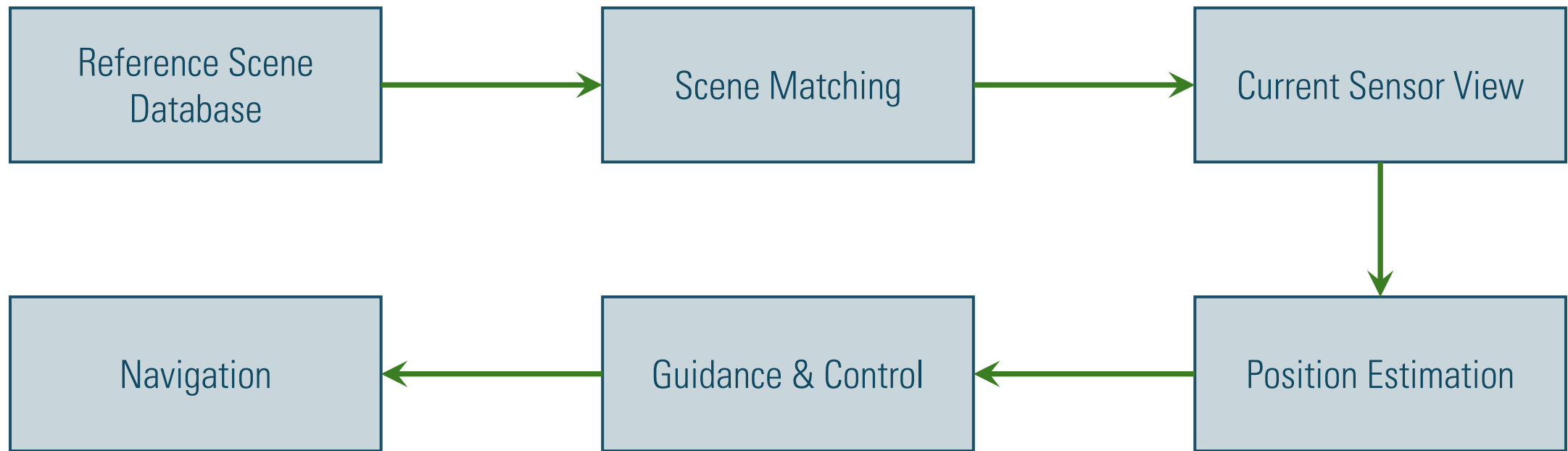
Why is DSMAC Important?

- Provides autonomous navigation without dependence on GPS.
- Enhances robustness in contested or signal-denied environments.
- Applicable to aerial vehicles, missiles, autonomous robots, and unmanned systems.
- Enables precise localization using onboard vision sensors.

Motivation

Modern navigation systems increasingly employ multiple sensing modalities, such as RGB and infrared (IR) cameras, to improve robustness under varying environmental and illumination conditions.

DSMAC NAVIGATION FRAMEWORK

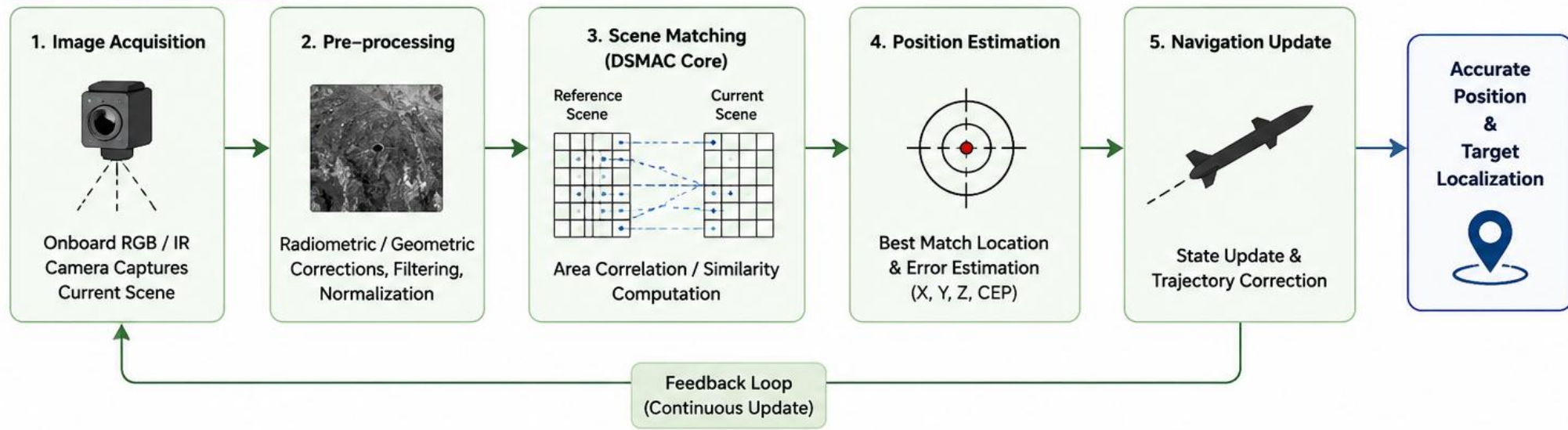


DSMAC – GENERAL WORKING PIPELINE

PRE-MISSION (OFFLINE)



IN-MISSION (ONLINE)



LITERATURE REVIEW

1. Original DSMAC (1980)

Reference

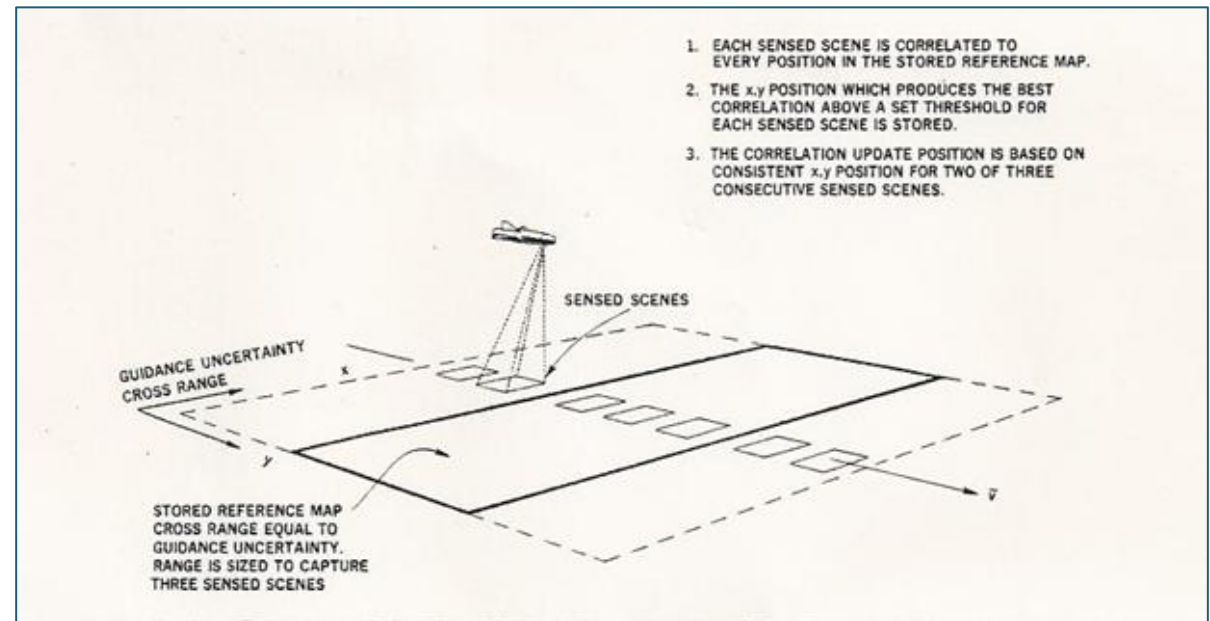
- Carr & Sobek, *Digital Scene Matching Area Correlator (DSMAC)*, Naval Avionics Center, 1980

Approach

- Area-based image correlation
- Correlation Surface Peak Detection
- Pixel-wise scene comparison
- DSMAC-assisted terminal guidance
- Reference image matching

Limitations

- Sensitive to illumination and viewpoint changes
- High computational complexity
- Limited adaptability



LITERATURE REVIEW

2. Advanced Scene Matching (1994)

Reference

- Irani & Christ, *Image Processing for Tomahawk Scene Matching*, Johns Hopkins APL Technical Digest, 1994

Approach

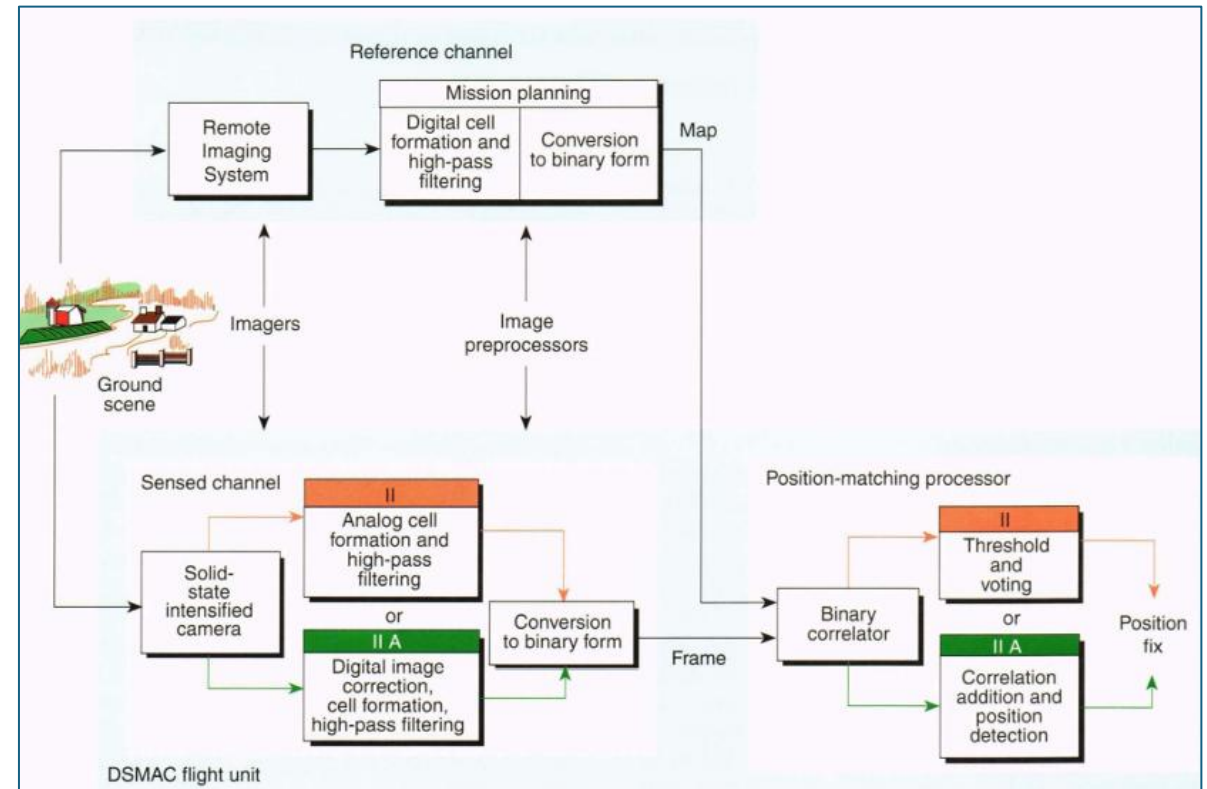
- Enhanced image processing techniques
- Improved scene registration
- Robust geometric alignment
- Optimized image matching for terminal guidance

Contributions

- Improved robustness over classical DSMAC
- Better handling of scene distortions
- Increased localization accuracy

Limitations

- Still relied on handcrafted image processing
- No learned feature representation
- Sensitive to environmental variations



LITERATURE REVIEW

3. AI-Based DSMAC (2026)

Reference

- Kashyap et al., *ML Based Solution for DSMAC System in Aerial Navigation*, 2026.
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Approach

- ORB feature extraction
- ResNet-50 feature learning
- FAISS similarity search
- CNN-based aerial image matching

Contributions

- Deep feature representation
- Faster large-scale image retrieval
- Improved robustness for aerial navigation
- Automated dataset generation

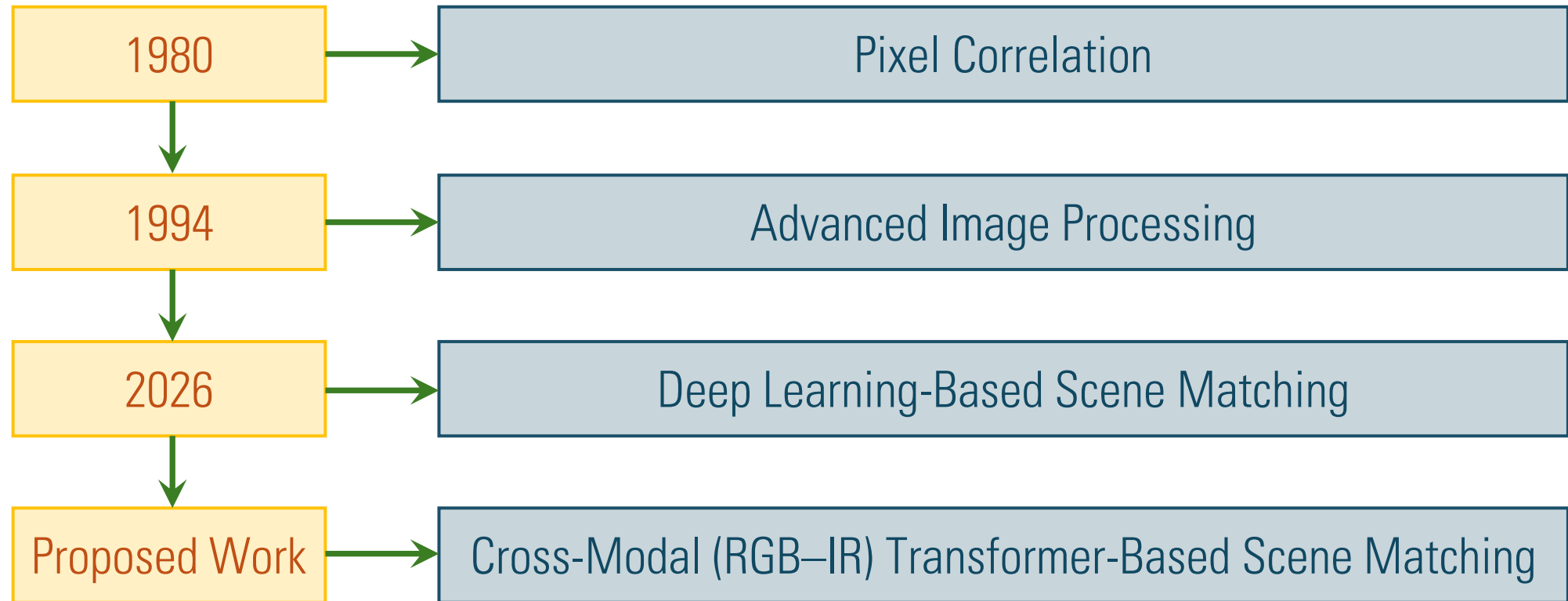
Limitations

- High computational requirements
- Limited multimodal (RGB–IR) capability

Table 5. Model Comparison Overview

Model	Accuracy	Processing Time	Strengths	Limitations
ORB	High	15 s	Accurate matching of keypoints	Too slow for real-time use
FAISS	Moderate	less than 1 s	Efficient, robust and scalable	Required pre-trained embeddings

EVOLUTION OF DSMAC IMAGE MATCHING



CURRENT STATE-OF-THE-ART FEATURE MATCHING METHODS

Method	Approach	Strength	Limitation
Superpoint + Superglue	Sparse keypoints + GNN matching	Accurate and robust matching	High computational cost; RGB-trained
LoFTR	Detector-free transformer- based dense matching	Excellent in low-texture regions	Computationally intensive; RGB-focused
Efficient LoFTR	Lightweight transformer- based dense matching	Faster inference with comparable accuracy	Limited validation on RGB- IR imagery

CURRENT STATE-OF-THE-ART FEATURE MATCHING METHODS (CONTD.)

Method	Approach	Strength	Limitation
MatchFormer	Coarse-to-fine hierarchical transformer	Robust multi-scale feature matching	High memory usage; not optimized for real-time
ASpanFormer	Adaptive span attention transformer	Handles large viewpoint changes	Complex architecture; higher inference latency
RoMa	Regression-based dense correspondence	Accurate dense pixel-wise matching	Computationally expensive; limited real-time deployment
LightGlue	Adaptive transformer feature matcher	Fast and efficient feature matching	Sparse matching; depends on extracted keypoints

COMPARATIVE ANALYSIS OF EXISTING IMAGE MATCHING METHODS

Method	Matching Strategy	Real-Time Performance	RGB–IR Support	Major Limitation
SIFT	Scale-Invariant Feature Matching	Low	No	Computationally expensive; handcrafted descriptors
SURF	Hessian-based Feature Matching	Moderate	No	Handcrafted features, RGB-focused
ORB	Keypoint Matching	High	No	Performance degrades in feature-poor regions
Classical DSMAC (1980)	Area Correlation	Low	No	Sensitive to illumination and viewpoint changes
Advanced DSMAC (1994)	Scene Registration	Moderate	No	Handcrafted processing, limited adaptability

COMPARATIVE ANALYSIS OF EXISTING IMAGE MATCHING METHODS (CONTD.)

Method	Matching Strategy	Real-Time Performance	RGB–IR Support	Major Limitation
SuperPoint	Keypoint Detection	High	No	Trained primarily on RGB datasets
SuperGlue	Graph Neural Network	Moderate	No	High computational complexity
LoFTR	Transformer-based Dense Matching	Moderate	No	Computationally intensive
EfficientLoFTR	Efficient Transformer	High	No	Limited evaluation on cross-modal imagery
LightGlue	Adaptive Transformer	High	No	Sparse correspondence, RGB-focused

LIMITATIONS OF EXISTING APPROACHES

Classical DSMAC

- Relies on pixel-level correlation, making it highly sensitive to illumination and viewpoint variations.
- Performance degrades significantly under dynamic environmental conditions.
- Computationally expensive for large search spaces.

Feature-Based Methods

- Depend on handcrafted feature descriptors.
- Reduced matching accuracy in texture-poor and repetitive environments.
- Limited robustness against large appearance changes.

Deep Learning-Based Methods

- Primarily trained and evaluated on RGB image datasets.
- Most methods assume same-modality image pairs during feature matching.
- Computational complexity limits deployment in high-frequency navigation systems.
- Limited evaluation under real-time embedded constraints.

RESEARCH GAP

Lack of **modality-invariant feature representations** for RGB–IR scene matching.

Absence of robust cross-modal correspondence estimation in existing DSMAC systems.

Need for dense feature matching with low computational latency.

Existing methods do not simultaneously satisfy **accuracy, robustness, and low-latency performance (50–100 Hz)** required for next-generation DSMAC.

PROPOSED METHODOLOGY

Objective

To develop a **multimodal DSMAC framework** capable of robust scene matching between RGB and infrared imagery by learning a shared latent feature space and performing efficient transformer-based feature correspondence.

Proposed Framework

Step 1: Multimodal Image Acquisition

- Simultaneous acquisition of RGB and Infrared (IR) images using synchronized onboard sensors.
- Image preprocessing including geometric alignment and normalization.

Step 2: Dual Encoder Network

- Independent feature encoders for RGB and IR images.
- Learn modality-specific representations while preserving structural scene information.
- Project both modalities into a **shared latent feature space**.

PROPOSED METHODOLOGY (CONTD.)

Step 3: Shared Latent Space Learning

- Align RGB and IR feature embeddings using a shared feature representation.
- Reduce modality discrepancy while preserving semantic correspondence.
- Generate modality-invariant descriptors suitable for feature matching.

Step 4: EfficientLoFTR-based Feature Matching

- Apply EfficientLoFTR to perform dense feature correspondence directly in the shared latent space.
- Utilize transformer-based attention mechanisms for robust matching.
- Eliminate dependency on handcrafted keypoint detectors.

Step 5: DSMAC Localization

- Estimate scene correspondences between the onboard image and reference database.
- Compute localization error.
- Update navigation state and trajectory correction.
- Repeat continuously for real-time guidance.

PROPOSED METHODOLOGY (CONTD.)

Expected Advantages

Robust RGB–IR feature correspondence.

Modality-invariant feature representation.

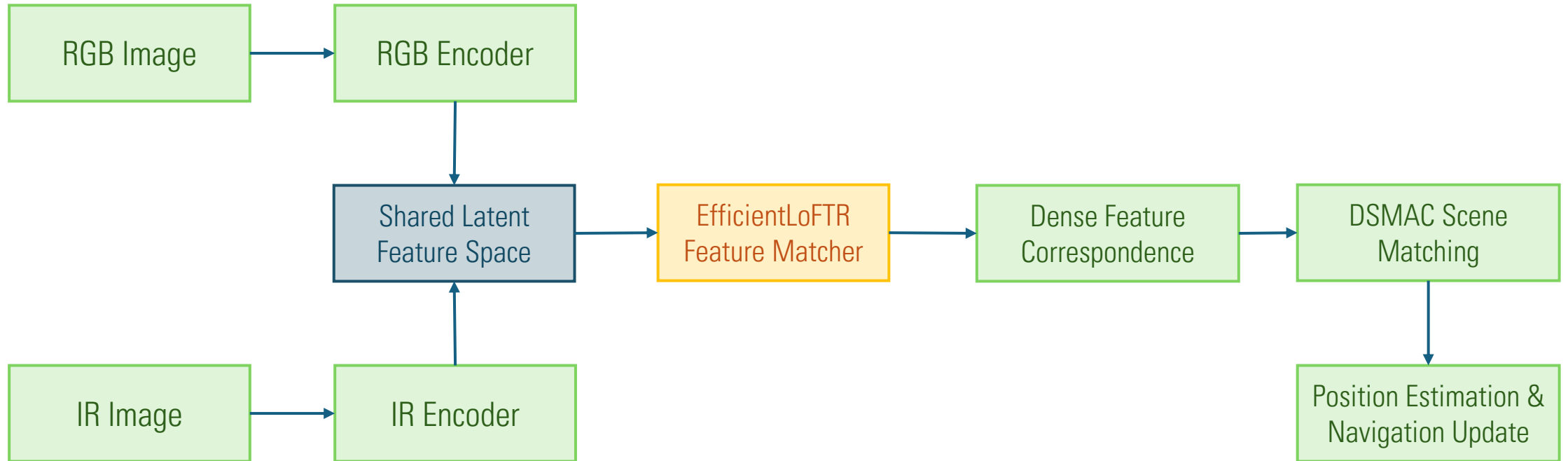
Reduced computational overhead using EfficientLoFTR.

Target low-latency performance of **50–100 Hz**.

Improved localization accuracy under varying illumination conditions.

Scalable architecture for integration with next-generation autonomous navigation systems.

PROPOSED METHODOLOGY (CONTD.)



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